Quantum Mechanics Problem Sheet

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1 Atomic Scale

1.1 Radius of an Atom and its Nucleus

The radius of an atom is around 10^{-10} m. The radius of a nucleus of an atom is around 10^{-15} m, which is around $100,000$ times smaller than the radius of the atom.

1.2 Atoms in a Marble vs. Marbles in the Earth

To find the number of atoms that can fit within a marble, we need to calculate the volume of an atom. The volume of a sphere is given by the formula:

$$\text{Volume of a sphere} = \frac{4}{3}\pi r^3$$

Substituting in the radius of an atom where $r \approx 1 \times 10^{-10}$m:

$$\text{volume of an atom} = \frac{4}{3}\pi(1 \times 10^{-10})^3 \approx 4.19 \times 10^{-30}$$

We can conclude that the order of magnitude of an atom is around 10^{-30} m^3. Next, we calculate the volume of a marble with diameter 3.6 cm , where $r \approx 1.8 \times 10^{-2}$m:

$$\text{Volume of a marble} = \frac{4}{3}\pi(1.8 \times 10^{-2}\ \text{m})^3 \approx 2.44 \times 10^{-5}\ \text{m}^3$$

Dividing the volume of the marble by the volume of an atom gives:

$$\text{Number of atoms in a marble} = \frac{2.44 \times 10^{-5}\ \text{m}^3}{10^{-30}\ \text{m}^3} = 2.44 \times 10^{25}$$

Therefore, there are around 10^{25} atoms in a marble with diameter 3.6 cm.

Answering the second part of the question requires calculating the volume of the Earth which has a diameter of 1.28×10^7m:

$$\text{Volume of Earth} = \frac{4}{3}\pi(0.64 \times 10^7)^3 \approx 1.1 \times 10^{21}\text{m}^3$$

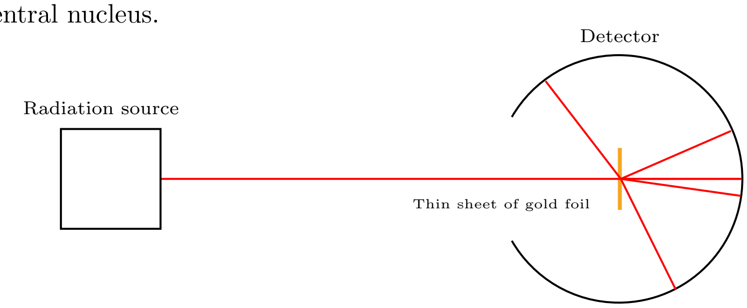
Dividing the volume of the Earth by the volume of a marble gives:

$$\text{Number of marbles in Earth} = \frac{1.10 \times 10^{21}\ \text{m}^3}{2.44 \times 10^{-5}\ \text{m}^3} \approx 4.5 \times 10^{25}$$

Therefore, around 10^{25} marbles can fit inside the Earth, which is the same order of magnitude as the number of atoms in a single marble. This real world example helps us understand how small atoms are.

2 Rutherford Experiment

The Rutherford scattering experiments, performed by Hans Geiger and Ernest Marsden between 1906 and 1913 provided evidence for the nuclear model of the atom, in which all the charge and most of the mass were concentrated in a tiny central nucleus.



A radioactive source that emitted alpha particles was focused into a narrow beam after passing through a slit in a sheet of lead. A thin sheet of gold leaf was placed in front of the beam, and a fluorescent screen coated with zinc sulfide surrounded it. When alpha particles hit the sheet, a small flash of light is produced, called a scintillation. A viewing microscope placed behind the screen allowed scientists to see the flash. During the experiment, most of the alpha particles passed straight through the gold foil, implying that atoms are made up mostly of empty space. Some particles were deflected slightly, suggesting that there were positively charged particles within the atoms interacting with the alpha radiation. However, some alpha particles were deflected at large angles and some were deflected straight back. This strong force of repulsion could only be accounted for by a heavy positively charged particle, the nucleus.

3 Black Body Radiation and Wien's Law

3.1 Temperature of a Tungsten Filament

To find the temperature of the tungsten filament lamp, we can use the Stefan-Boltzmann law

$$P = e\sigma AT^4$$

where power is equal to the product of the emissivity of the material, its surface area and the Stefan-Boltzmann constant ($5.67 \times 10^{-8}\ \text{W/m}^2\text{K}^4$).

The surface area of the filament:

$$A = 0.2 \times 0.14 \times 10^{-3} = 2.8 \times 10^{-5} m^2$$

The temperature of the filament is given by the formula:

$$T = \left(\frac{P}{e\sigma A}\right)^{1/4}$$

When the input power is 100W and the tungsten filament has a surface area of 2.8×10^{-5} and an emissivity of 0.3:

$$T = \left(\frac{100}{0.3 \times 5.67 \times 10^{-8} \times 2.8 \times 10^{-5}}\right)^{1/4} = 3806.58\text{K}$$

$$3806.58 - 273.15 = 3533.4°\text{C}$$

This is higher than the melting point of tungsten, which is 3695K. We have assumed that 100W is the radiant power, which may not be the case since it emits power across a large range of the EM spectrum. Humans can only see the 400nm to 700nm range of the spectrum, which means we do not see 100W of power. We also assume that light is only emitted from one side of the filament, but in reality it radiates from its whole surface. It is unlikely that the filament emits 100W of visible light, and so our assumption is unreasonable.

3.2 Luminosity of the Sun

Luminosity is defined as the total radiant power of a star. The Sun's luminosity can be calculated using the standard Stefan-Boltzmann law but with emissivity set to 1, as the Sun closely approximates a black body; a theoretical object that absorbs all incident EM radiation and re-radiates it. The surface area of a sphere is given:

$$4\pi r^2$$

Therefore, the formula for luminosity is:

$$L = 4\pi r^2 \sigma T^4$$

The luminosity of the sun where $r = 695,500,000$m and the surface temperature is 5778K is given:

$$L = 4\pi \times 695,500,000^2 \times 5.67 \times 10^{-8} \times 5778^4 = 3.84 \times 10^{26}\text{W}$$

3.3 Solar Panel Size on Mars

Calculating the width of a solar panel which delivers 100W of power on the surface, first requires us to account for its 20% efficiency. Therefore the required input power is:

$$P_{required} = 100 \times \frac{1}{0.2} = 500$$

W

Next, we need to calculate the intensity of solar radiation on the surface of Mars using the inverse square law where intensity of solar radiation is inversely proportional to the square of the distance from the source, since light spreads out in all directions:

$$I_{Mars} = \frac{L}{4\pi r^2}$$

Intensity on the surface of Mars, which is 227.9 million km from the Sun:

$$I_{Mars} = \frac{3.84 \times 10^{26}\text{W}}{4\pi \times (227.9 \times 10^6\text{m})^2} = 588.35\text{W/m}^2$$

To calculate the area of the solar panel required:

$$A = \frac{P}{I}$$

$$A = \frac{500\text{W}}{588.35\text{W/m}^2} = 0.8498\text{m}^2$$

$$width = \sqrt{0.8498} = 0.92\text{m}$$

3.4 Radiation from Human Skin

Calculating the wavelength of radiation given off by human skin requires us to use Wien's displacement law, which describes the inverse relationship between the peak wavelength emitted from an object and its temperature, where b is Wien's constant:

$$\lambda_{max} = \frac{b}{T} \text{ where } b = 2.898 \times 10^{-3}\text{mK}$$

The surface skin temperature of a human is around 34°C. In Kelvin:

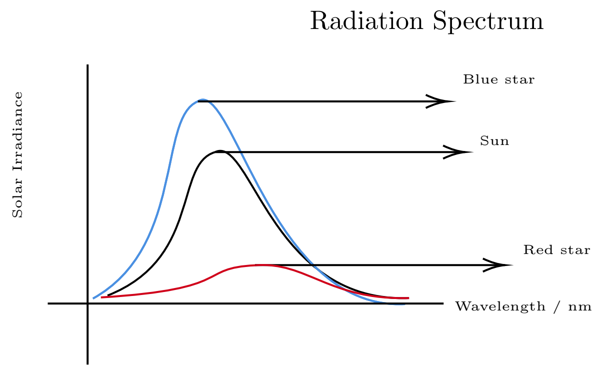
$$T = 34 + 273.15 = 307.15\text{K}$$

Calculating the maximum wavelength:

$$\lambda_{max} = \frac{2.898 \times 10^{-3}\text{mK}}{307.15\text{K}} = 9.435 \times 10^{-6}\text{m} = 9435\text{nm}$$

This peak wavelength lies within the infrared region of the electromagnetic spectrum, which explains why IR cameras can be used to detect heat signatures of humans and other warm-blooded animals.

3.5 Radiation Spectrum Comparison



Calculating the maximum wavelength of radiation given off by the sun requires utilising Wien's displacement law again, where the surface temperature of the sun is 5778K.

$$\lambda_{max} = \frac{2.898 \times 10^{-3}\text{mK}}{5778\text{K}} = 5.01 \times 10^{-7}\text{m} = 502\text{nm}$$

3.6 Expansion of the Sun into a Red Giant

Calculating the expansion of the solar radius when $\lambda_{max} = 680$nm first requires us to first find the current peak wavelength emitted by the sun using Wien's displacement law. We have already calculated this in the previous part, giving us a wavelength of 502nm.

Next, we need to calculate the surface temperature of the red giant using Wien's displacement law:

$$T_{\text{red giant}} = \frac{2.898 \times 10^{-3} \text{ m K}}{680 \times 10^{-9} \text{ m}} = 4265 \text{ K}$$

Since luminosity is constant we can equate the luminosity of the Sun and the Red Giant:

$$4\pi R_{sun}^2 \sigma T_{sun}^4 = 4\pi R_{\text{red giant}}^2 \sigma T_{\text{red giant}}^4$$

We can cancel out $4\pi\sigma$ to find the radius ratio:

$$\frac{R_{\text{red giant}}}{R_{sun}} = \left(\frac{T_{sun}}{T_{\text{red giant}}}\right)^2$$

Substituting in our temperature values gives us the expansion ratio:

$$\frac{R_{\text{red giant}}}{R_{sun}} = \left(\frac{5778}{4265}\right)^2 = 1.84$$

The Sun's radius expands to approximately 1.84x its current size as it becomes a red giant.

4 Photoelectric Effect

4.1 Maximum Wavelength for Photoemission from Silver

The equation relating the work function of a material and the maximum kinetic energy of emitted electrons is given:

$$KE_{max} = hf - \phi$$

To determine the maximum wavelength that will just enable photoelectrons to be emitted from the surface of silver,we set $KE_{max} = 0$. This simplifies the equation to:

$$\phi = hf$$

Frequency and wavelength of light are related by the equation $c = f\lambda$, where c is the speed of light. We can rewrite the equation as:

$$\frac{hc}{\lambda_{max}} = \phi$$

Maximum wavelength is given by:

$$\lambda_{max} = \frac{hc}{\phi}$$

Silver has a work function of 4.3 eV, where 1 eV $= 1.6 \times 10^{-19}$J. The maximum wavelength is given:

$$\lambda_{max} = \frac{hc}{\phi} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{4.3 \times 1.6 \times 10^{-19}} = 2.89 \times 10^{-7}\text{m} = 289\text{nm}$$

This wavelength lies in the ultraviolet part of the EM spectrum, which means that silver will only emit photoelectrons when illuminated by UV light or shorter wavelengths.

4.2 Energy of Emitted Photoelectrons

Calculating the energy of photoelectrons produced, when UV light with a wavelength of 200nm is shone onto silver with a work function of 4.3eV is given:

$$KE_{max} = \frac{hc}{\lambda} - \phi$$

$$E = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{200 \times 10^{-9}} - 4.3 \times 1.6 \times 10^{-19} = 3.065 \times 10^{-19}\text{J}$$

4.3 Photon Absorption Rate

Finding the number of photons absorbed by silver per second, where the power of the incident light is 1μW is calculated by dividing the energy incident on the silver by the energy of each individual photon:

$$\text{number of photons per second} = \frac{Pt}{\frac{hc}{\lambda}} = \frac{1 \times 10^{-6}}{\frac{6.63\times10^{-34}\times3\times10^8}{200\times10^{-9}}} = 1.01\times10^{12} \text{ photons per second}$$

4.4 Photocurrent

One photoelectron is produced per absorbed UV photon. This means the rate of photoelectron emission equals the rate of photon absorption calculated in the previous part:

$$1.01 \times 10^{12} \times 1.6 \times 10^{-19} = 1.61 \times 10^{-7}\text{A} = 161\text{nA}$$

Current is defined as the rate of flow of charge:

$$I = \frac{Q}{t}$$

Since each photoelectron carries a charge e, the charge produced per second is the number of electrons per second multiplied by e. As the electron rate is already expressed *per second*, this directly gives the current:

$$I = (\text{number of electrons per second}) \times e$$

$$I = 1.01 \times 10^{12} \times 1.6 \times 10^{-19} = 1.61 \times 10^{-7}\text{A} = 161\text{nA}$$

4.5 Stopping Voltage

Finding the stopping voltage required to reduce the photoelectric current to zero, requires us to equate the maximum kinetic energy of the emitted photons to electric potential energy required to stop them:

$$V = \frac{KE_{max}}{e}$$

$$V = \frac{3.065 \times 10^{-19}}{1.6 \times 10^{-19}} = 1.92\text{V}$$

5 Electron Diffraction and the de Broglie Wavelength

5.1 Velocity of an Accelerated Electron

To find the final velocity we can use the formula for kinetic energy, where the kinetic energy is the work done by the accelerating p.d., and m is the mass of the electron:

$$KE = \frac{1}{2}mv^2$$

$$eV = \frac{1}{2}mv^2$$

$$\frac{2eV}{m} = v^2$$

$$v = \sqrt{\frac{2eV}{m}}$$

5.2 Relativistic Limits

To investigate the relativistic effects of accelerating electrons, we can setup an inequality where the speed of electrons is less than 1% the speed of light:

$$m = \text{mass of an electron} = 9.109 \times 10^{-31}\text{kg}$$

$$c = \text{speed of light} = 2.998 \times 10^{8}\text{ms}^{-1}$$

$$v < 0.01c$$

$$\sqrt{\frac{2eV}{m}} < 0.01c$$

$$V < \frac{0.01^2 \times mc^2}{2e}$$

$$V < 25.58 \text{ volts}$$

We can therefore ignore relativistic effects, since $v << c$. Furthermore, the required voltage to reach 0.01c is not that high at 25.6v. If we look at an electron diffraction practical, where the accelerating p.d. is around 5kV:

$$v = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 5000}{9.109 \times 10^{-31}}} = 41910674 \text{ ms}^{-1}$$

$$\frac{41910674}{2.998 \times 10^8} = 0.14c$$

At this p.d. the speed of the electrons is 14% the speed of the light, which means we need to start accounting for relativistic effects. In particle accelerators, where electrons are accelerated at millions of volts, the classical formula will no longer work and we must instead use the relativistic equation to cover the effects:

$$eV = (\gamma - 1)mc^2$$

5.3 Derivation of the de Broglie Wavelength

The formula for the de-Broglie wavelength of an electron can be derived as follows:

$$p = mv$$

$$KE = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$eV = \frac{p^2}{2m}$$

$$p = \sqrt{2meV}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$$

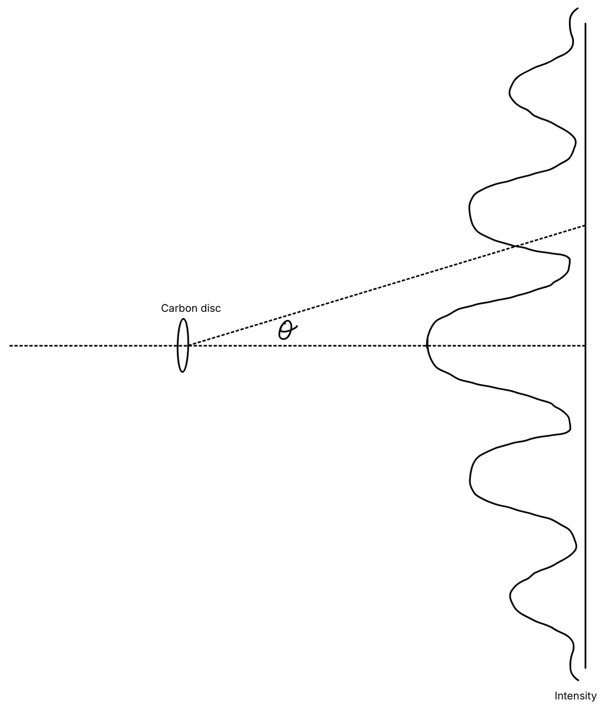


Figure 1: Intensity against diffraction angle

5.4 Electron Diffraction Pattern

The pattern shown above by the graph is an interference pattern. This can be explained by the dual wave-particle nature of electrons, when accelerated fast enough. They behave as waves with a de-Broglie wavelength similar in magnitude to the spacing between the carbon atoms.

5.5 Accelerating Voltage for the First Maximum

To find the accelerating p.d. required to achieve the first maxima at 10°, we can use the diffraction grating equation, where d is the atomic spacing in the carbon disc:

$$n\lambda = dsin\theta$$

Since, we are looking at the first peak, n = 1.

$$\lambda = 1.23 \times 10^{-10} \times sin(10^\circ) = 2.14 \times 10^{-11}\text{m}$$

To find the accelerating p.d. we can use the equation we derived earlier:

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$V = \frac{h^2}{2me\lambda^2}$$

$$V = \frac{(6.63 \times 10^{-34})^2}{2 \times 9.109 \times 10^{-31} \times 1.6 \times 10^{-19} \times (2.14 \times 10^{-11})^2} = 3292\text{V} = 3.3\text{kV}$$

5.6 Angles of Higher-Order Maxima

To create a more accurate graph for intensity against diffraction angle, we need to calculate the exact angles at which the peaks occur. We also need to find out how many peaks there are. We can solve both of these using the diffraction grating formula from earlier:

$$\text{Max number of peaks} = \frac{dsin(90^\circ)}{\lambda} = \frac{1.23 \times 10^{-10} \times sin(90^\circ)}{2.14 \times 10^{-11}} = 5.75$$

The number of peaks must be a whole number, so the maximum on our graph will be 5 peaks.

To find the diffraction angles at which these occur, we just solve the equation for θ, where n is the respective number of the peak.

$$\theta = \arcsin\left(\frac{n\lambda}{d}\right)$$

Below is a table of data of all the peaks.

n	-5	-4	-3	-2	-1	0	1	2	3	4	5
$\theta/^\circ$	-60.45	-44.10	-31.46	-20.36	-10.02	0	10.02	20.36	31.46	44.10	60.45

Below is a sketch of a more accurate graph:

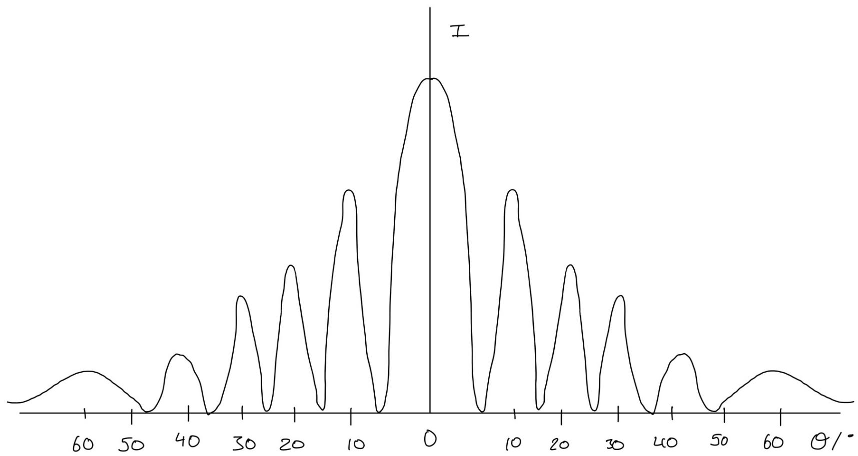


Figure 2: Improved sketch including all five orders of diffraction maxima

6 Hydrogen Spectral Lines

The wavelength of the photon emitted when n=3 in the Lyman series can be calculated using the Rydberg Formula where R_H is the Rydberg constant, n_2 is higher energy level, and n_1 is the lower energy level:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{1^2} - \frac{1}{3^2} \right) = 1.097 \times 10^7 \times \frac{8}{9} = 9.751 \times 10^6 \text{ m}^{-1}$$

$$\lambda = \frac{1}{9.751 \times 10^6} = 1.025 \times 10^{-7}\text{m} = 102.5 \text{ nm}$$

This lies in the UV light region of the EM spectrum.
To calculate it for the Balmer and Pfund line, we can follow the same method.
For the Balmer line where the transition is from E_4 to E_2:

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = 1.097 \times 10^7 \times \frac{3}{16} = 2.057 \times 10^6 \text{ m}^{-1}$$

$$\lambda = \frac{1}{2.057 \times 10^6 \text{ m}^{-1}} = 4.861 \times 10^{-7} \text{ m} = 486.1 \text{ nm}$$

This lies in the visible light region of the EM spectrum, specifically the colour cyan.

For the Pfund line where the transition is from E_7 to E_5:

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{5^2} - \frac{1}{7^2}\right) = 1.097 \times 10^7 \times \left(\frac{1}{25} - \frac{1}{49}\right) \approx 2.150 \times 10^5 \text{ m}^{-1}$$

$$\lambda = \frac{1}{2.150 \times 10^5 \text{ m}^{-1}} \approx 4.651 \times 10^{-6} \text{ m} = 4651 \text{ nm}$$

This lies in the infrared region of the EM spectrum.